Assignment No: 2

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Subject: DSML

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Housing.csv")

print("=" * 60)
print("FIRST FIVE RECORDS")
print("=" * 60)
print(df.head())
```

O/p:

```
...  ============================================================
     FIRST FIVE RECORDS
     ============================================================
           price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
     0  13300000  7420         4          2        3      yes        no       no
     1  12250000  8960         4          4        4      yes        no       no
     2  12250000  9960         3          2        2      yes        no      yes
     3  12215000  7500         4          2        2      yes        no      yes
     4  11410000  7420         4          1        2      yes       yes      yes

       hotwaterheating airconditioning  parking prefarea furnishingstatus
     0              no             yes        2      yes        furnished
     1              no             yes        3       no        furnished
     2              no              no        2      yes   semi-furnished
     3              no             yes        3      yes        furnished
     4              no             yes        2       no        furnished
```

Code:

```
print("\n" + "=" * 60)
```

```
print("DATASET INFORMATION")
print("=" * 60)
print(df.info())

print("\n" + "=" * 60)
print("DATASET SHAPE")
print("=" * 60)
print("Rows and Columns :", df.shape)

print("\n" + "=" * 60)
print("MISSING VALUES")
print("=" * 60)
print(df.isnull().sum())
```

O/p:

```
...
    ============================================================
    DATASET INFORMATION
    ============================================================
    <class 'pandas.core.frame.DataFrame'>
    RangeIndex: 545 entries, 0 to 544
    Data columns (total 13 columns):
     #   Column            Non-Null Count  Dtype
    ---  ------            --------------  -----
     0   price             545 non-null    int64
     1   area              545 non-null    int64
     2   bedrooms          545 non-null    int64
     3   bathrooms         545 non-null    int64
     4   stories           545 non-null    int64
     5   mainroad          545 non-null    object
     6   guestroom         545 non-null    object
     7   basement          545 non-null    object
     8   hotwaterheating   545 non-null    object
     9   airconditioning   545 non-null    object
     10  parking           545 non-null    int64
     11  prefarea          545 non-null    object
     12  furnishingstatus  545 non-null    object
    dtypes: int64(6), object(7)
    memory usage: 55.5+ KB
    None
```

```
...
    ============================================================
    DATASET SHAPE
    ============================================================
    Rows and Columns : (545, 13)

    ============================================================
    MISSING VALUES
    ============================================================
    price              0
    area               0
    bedrooms           0
    bathrooms          0
    stories            0
    mainroad           0
    guestroom          0
    basement           0
    hotwaterheating    0
    airconditioning    0
    parking            0
    prefarea           0
    furnishingstatus   0
    dtype: int64
```

Code:

```
print("\n" + "=" * 60)
print("SUMMARY STATISTICS")
print("=" * 60)

numeric_columns = ['price', 'area']

for column in numeric_columns:

    print("\nColumn :", column)
    print("----------------------------")
    print("Mean             :", round(df[column].mean(), 2))
    print("Median           :", round(df[column].median(), 2))
    print("Standard Deviation:", round(df[column].std(), 2))
    print("Variance         :", round(df[column].var(), 2))

print("\nComplete Statistical Summary")
print(df.describe())
```

O/p:

```
============================================================
SUMMARY STATISTICS
============================================================

Column : price
------------------------------
Mean              : 4766729.25
Median            : 4340000.0
Standard Deviation: 1870439.62
Variance          : 3498544355820.58

Column : area
------------------------------
Mean              : 5150.54
Median            : 4600.0
Standard Deviation: 2170.14
Variance          : 4709512.06

Complete Statistical Summary
              price          area    bedrooms   bathrooms     stories  \
count  5.450000e+02    545.000000  545.000000  545.000000  545.000000
mean   4.766729e+06   5150.541284    2.965138    1.286239    1.805505
std    1.870440e+06   2170.141023    0.738064    0.502470    0.867492
min    1.750000e+06   1650.000000    1.000000    1.000000    1.000000
25%    3.430000e+06   3600.000000    2.000000    1.000000    1.000000
50%    4.340000e+06   4600.000000    3.000000    1.000000    2.000000
75%    5.740000e+06   6360.000000    3.000000    2.000000    2.000000
max    1.330000e+07  16200.000000    6.000000    4.000000    4.000000

          parking
count  545.000000
mean     0.693578
std      0.861586
min      0.000000
25%      0.000000
50%      0.000000
75%      1.000000
max      3.000000
```

Code:

```
plt.figure(figsize=(8,5))
plt.hist(df['price'],
        bins=20,
        color='skyblue',
        edgecolor='black')

plt.title("Histogram of Housing Price")
plt.xlabel("price")
plt.ylabel("Frequency")
plt.grid(True)
```

```
plt.show()

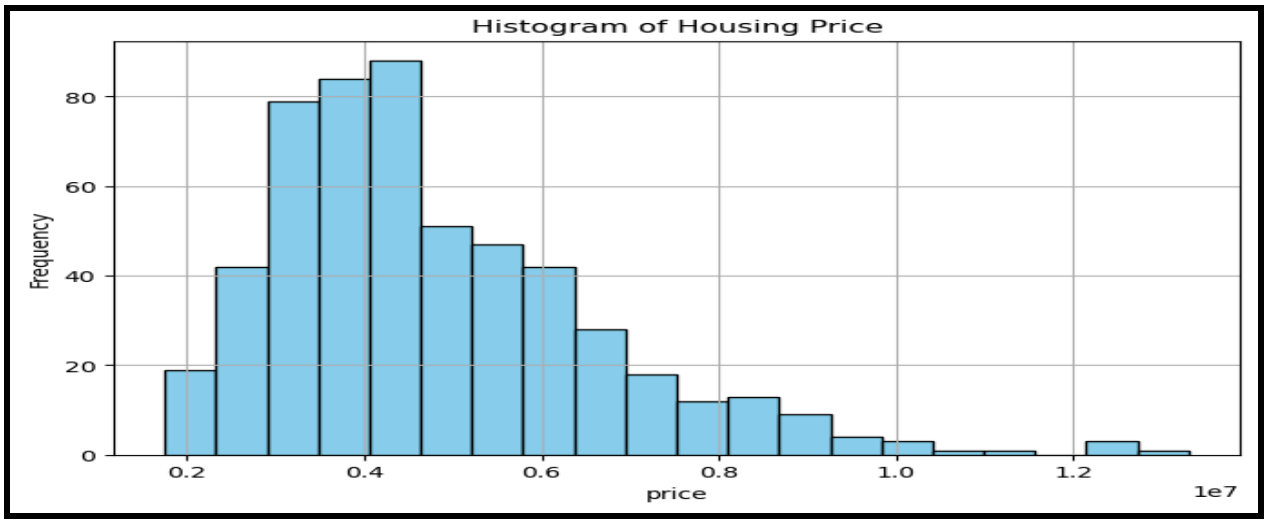
plt.figure(figsize=(8,5))
plt.hist(df['area'],
        bins=20,
        color='skyblue',
        edgecolor='black')

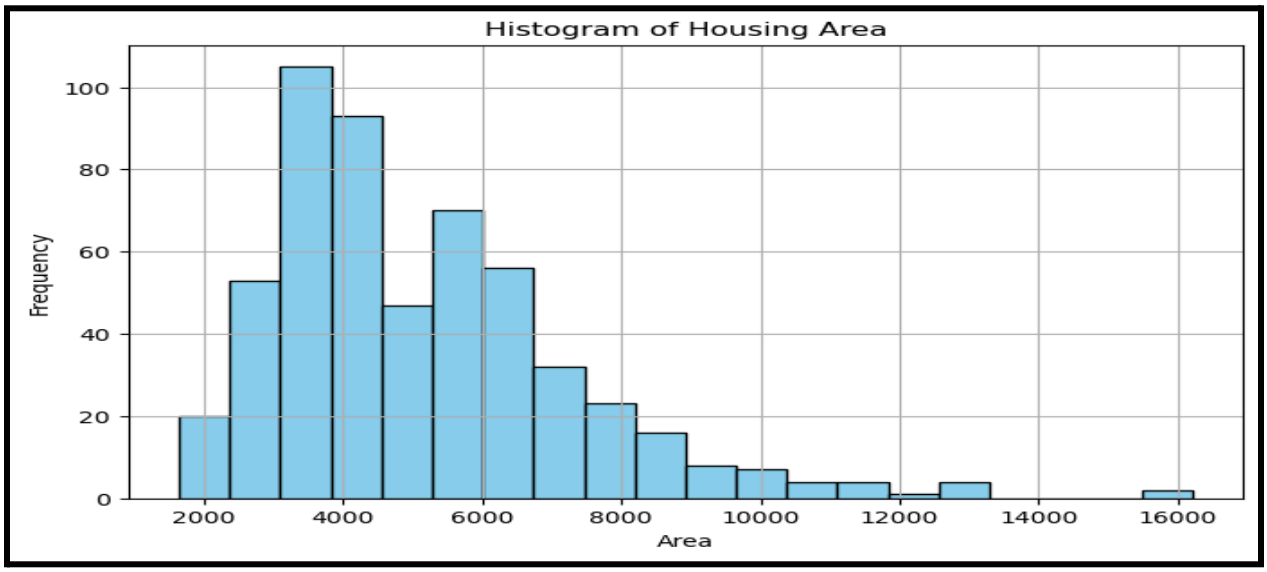
plt.title("Histogram of Housing Area")
plt.xlabel("Area")
plt.ylabel("Frequency")
plt.grid(True)
plt.show()

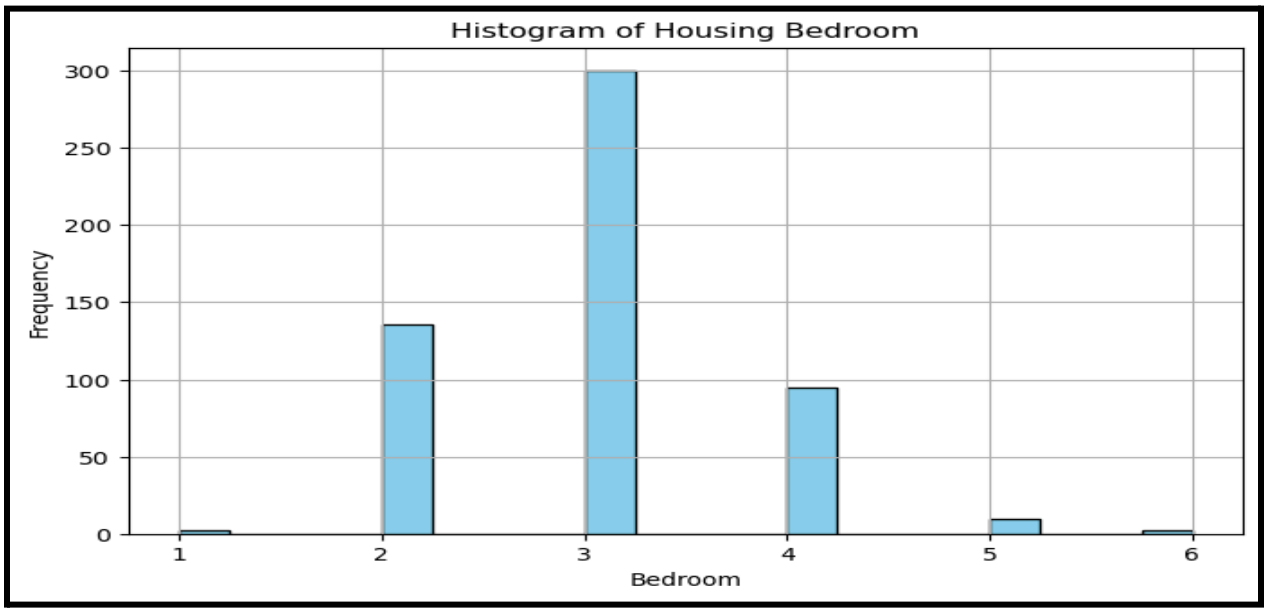
plt.figure(figsize=(8,5))
plt.hist(df['bedrooms'],
        bins=20,
        color='skyblue',
        edgecolor='black')

plt.title("Histogram of Housing Bedroom")
plt.xlabel("Bedroom")
plt.ylabel("Frequency")
plt.grid(True)
plt.show()
```

O/p:


Histogram of Housing Price
Frequency
price
1e7
0
20
40
60
80
0.2
0.4
0.6
0.8
1.0
1.2


Histogram of Housing Area
Frequency
Area
0
20
40
60
80
100
2000
4000
6000
8000
10000
12000
14000
16000


Histogram of Housing Bedroom
Frequency
Bedroom
0
50
100
150
200
250
300
1
2
3
4
5
6

Code:

```
import matplotlib.pyplot as plt

plt.figure(figsize=(5,8))

sns.boxplot(y=df['price'], color='skyblue')
plt.title("Box plot of Housing price")
plt.ylabel('price')

plt.grid(True)
plt.show()

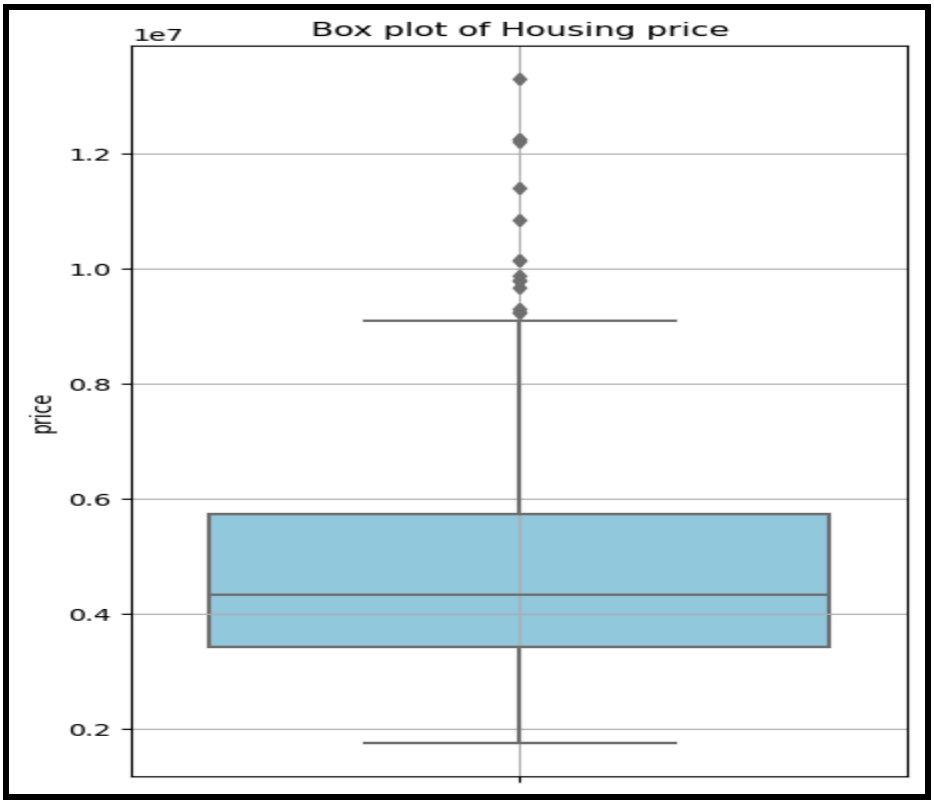
import matplotlib.pyplot as plt

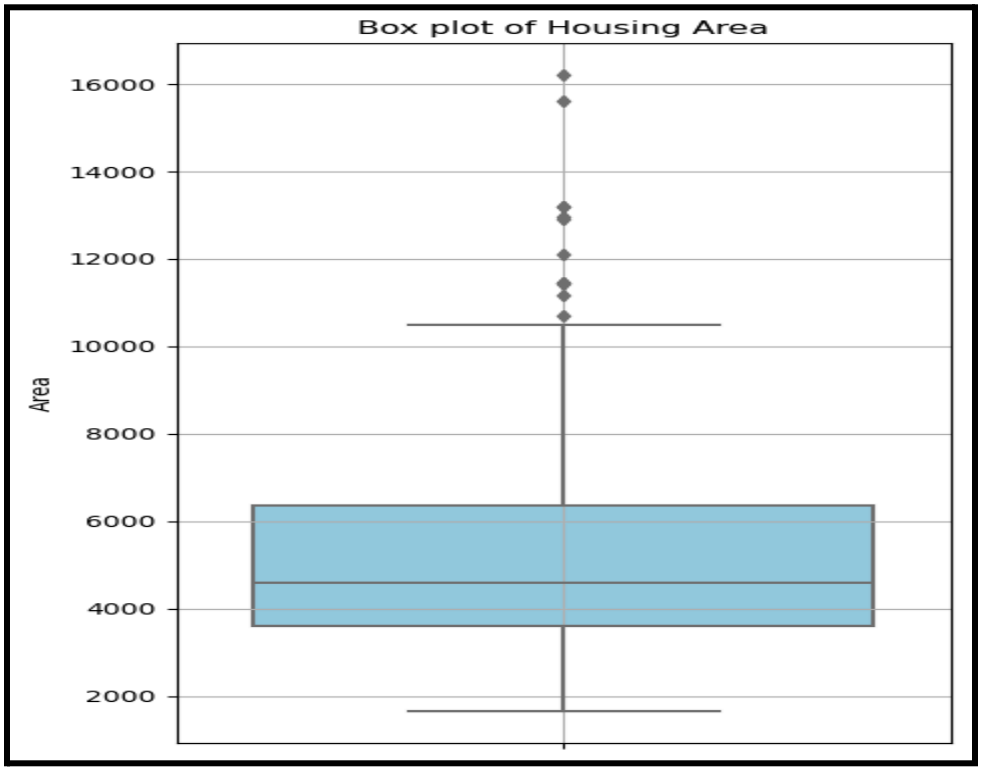
plt.figure(figsize=(5,8))

sns.boxplot(y=df['area'], color='skyblue')
plt.title("Box plot of Housing Area")
plt.ylabel('Area')

plt.grid(True)
plt.show()
```

O/p:


Box plot of Housing price
1e7
price
1.2
1.0
0.8
0.6
0.4
0.2


Box plot of Housing Area
Area
16000
14000
12000
10000
8000
6000
4000
2000